

EE5203 L02 Pre-Lab Quiz

Decisions, loops, and the eight-sample pipeline - Basri Erdogan

Preparation time: 5-8 minutes **Attempts:** Two; highest score recorded **Bring to the lab:** Your answer to Question 5

1. After these statements, what is the value of `x`?

```
int x = 3;
x = x + 2;
x++;
```

- a. 3
- b. 5
- c. 6
- d. 7

2. Which line stores a value, and which line asks a question?

```
status = 1;      /* line 1 */
status == 1     /* line 2 */
```

- a. Both store a value.
- b. Line 1 stores; line 2 compares without changing `status`.
- c. Line 1 compares; line 2 stores.
- d. Both compare.

3. How many times does this loop body run, and what is the last value `i` takes **inside** the body?

```
for (int i = 0; i < 8; i++) { /* body */ }
```

- a. 8 times; last inside value 7
- b. 8 times; last inside value 8
- c. 9 times; last inside value 8
- d. 7 times; last inside value 7

4. `uint16_t samples[8]` is declared. Which access is invalid?

- a. `samples[0]`
- b. `samples[7]`
- c. `samples[8]`
- d. `samples[3]`

5. Complete this entry ticket before class. The lab data is {850, 1250, 4095, 4200, 2050, 900, 3100, 2800}; values above 4095 are clamped to 4095; a high sample is strictly greater than 2500.

Prediction	Your answer
Which input value will be clamped, and to what?	
How many samples count as high after clamping?	
Integer result of <code>19140 / 8</code>	
First check if a <code>while</code> loop never terminates	